# MATHEMATICS AND STATISTICS

**Time: 3 Hrs.** **Max. Marks: 80**

***General instructions:***

*The question paper is divided into **FOUR** sections.*

*(1)* ***Section A:*** *Q. 1 contains **Eight** multiple choice type of questions, each carrying **Two** marks. Q. 2 contains **Four** very short answer type questions, each carrying **one** mark.*

*(2)* ***Section B:*** *Q. 3 to Q. 14 contain **Twelve** short answer type questions, each carrying **Two** marks. (Attempt any **Eight**)*

*(3)* ***Section C:*** *Q. 15 to Q. 26 contain **Twelve** short answer type questions, each carrying **Three** marks. (Attempt any **Eight**)*

*(4)* ***Section D:*** *Q. 27 to Q. 34 contain **Eight** long answer type questions, each carrying **Four** marks. (Attempt any **Five**)*

*(5) Use of log table is allowed. Use of calculator is not allowed.*

*(6) Figures to the right indicate full marks.*

*(7) Use of graph paper is <u>not</u> necessary. Only rough sketch of graph is expected.*

*(8) For each multiple choice type of question, it is mandatory to write the correct answer along with its alphabet, e.g. (a)…… / (b)…… /(c)…… /(d)……, etc. No marks shall be given, if <u>ONLY</u> the correct answer or the alphabet of correct answer is written. Only the first attempt will be considered for evaluation.*

*(9) Start answer to each section on a new page.*

## SECTION – A

**Q.1. Select and write the correct answer for the following multiple choice type of questions:** **[16]**

i. If $p \wedge q$ is F, $p \rightarrow q$ is F then the truth values of p and q are _____ respectively.

(a) T, T (b) T, F (c) F, T (d) F, F (2)

ii. In $\Delta ABC$, if $c^2 + a^2 - b^2 = ac$, then $\angle B$ = _____.

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$ (2)

iii. The area of the triangle with vertices (1, 2, 0), (1, 0, 2) and (0, 3, 1) in sq. unit is _____.

(a) $\sqrt{5}$ (b) $\sqrt{7}$ (c) $\sqrt{6}$ (d) $\sqrt{3}$ (2)

iv. If the corner points of the feasible solution are (0, 10), (2, 2) and (4, 0) then the point of minimum $z = 3x + 2y$ is _____.

(a) (2, 2) (b) (0, 10) (c) (4, 0) (d) (3, 4) (2)

v. If $y$ is a function of $x$ and $\log(x + y) = 2xy$, then the value of $y'(0)$ = _____.

(a) 2 (b) 0 (c) –1 (d) 1 (2)

vi. $\int \cos^3 x \, dx$ = _____.

(a) $\frac{1}{12}\sin 3x + \frac{3}{4}\sin x + c$ (b) $\frac{1}{12}\sin 3x + \frac{1}{4}\sin x + c$

(c) $\frac{1}{12}\sin 3x - \frac{3}{4}\sin x + c$ (d) $\frac{1}{12}\sin 3x - \frac{1}{4}\sin x + c$ (2)

vii. The solution of the differential equation $\frac{dx}{dt} = \frac{x \log x}{t}$ is _____

(a) $x = e^{ct}$ (b) $x + e^{ct} = 0$

(c) $x = e^t + t$ (d) $xe^{ct} = 0$ (2)

viii. Let the probability mass function (p.m.f.) of a random variable X be $P(X = x) = {}^4C_x \left(\frac{5}{9}\right)^x \times \left(\frac{4}{9}\right)^{4-x}$, for $x = 0, 1, 2, 3, 4$ then E(X) is equal to _____

(a) $\frac{20}{9}$ (b) $\frac{9}{20}$ (c) $\frac{12}{9}$ (d) $\frac{9}{25}$ (2)

**Q.2. Answer the following questions:** **[4]**

i. Write the joint equation of co-ordinate axes. (1)

ii. Find the values of c which satisfy $|c\bar{u}| = 3$ where $\bar{u} = \hat{i} + 2\hat{j} + 3\hat{k}$. (1)

iii. Write $\int \cot x \, dx$. (1)

iv. Write the degree of the differential equation $e^{\frac{dy}{dx}} + \frac{dy}{dx} = x$ (1)

## SECTION – B

**Attempt any EIGHT of the following questions:** **[16]**

**Q.3.** Write inverse and contrapositive of the following statement:
If $x < y$ then $x^2 < y^2$ (2)

**Q.4.** If $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ is a non singular matrix, then find $A^{-1}$ by elementary row transformations.

Hence write the inverse of $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (2)

**Q.5.** Find the cartesian co-ordinates of the point whose polar co-ordinates are $\left(\sqrt{2}, \frac{\pi}{4}\right)$. (2)

**Q.6.** If $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines and $h^2 = ab \neq 0$ then find the ratio of their slopes. (2)

**Q.7.** If $\bar{a}, \bar{b}, \bar{c}$ are the position vectors of the points A, B, C respectively and $5\bar{a} + 3\bar{b} - 8\bar{c} = \bar{0}$ then find the ratio in which the point C divides the line segment AB. (2)

**Q.8.** Solve the following inequations graphically and write the corner points of the feasible region:
$2x + 3y \leq 6, x + y \geq 2, x \geq 0, y \geq 0$ (2)

**Q.9.** Show that the function $f(x) = x^3 + 10x + 7, x \in R$ is strictly increasing. (2)

**Q.10.** Evaluate: $\int_0^{\frac{\pi}{2}} \sqrt{1 - \cos 4x} \, dx$ (2)

**Q.11.** Find the area of the region bounded by the curve $y^2 = 4x$, the X-axis and the lines $x = 1$, $x = 4$ for $y \geq 0$. (2)

**Q.12.** Solve the differential equation
$\cos x \cos y \, dy - \sin x \sin y \, dx = 0$ (2)

**Q.13.** Find the mean of number randomly selected from 1 to 15. (2)

**Q.14.** Find the area of the region bounded by the curve $y = x^2$ and the line $y = 4$. (2)

## SECTION – C

**Attempt any EIGHT of the following questions:** **[24]**

**Q.15.** Find the general solution of $\sin\theta + \sin 3\theta + \sin 5\theta = 0$ (3)

**Q.16.** If $-1 \le x \le 1$, the prove that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ (3)

**Q.17.** If $\theta$ is the acute angle between the lines represented by $ax^2 + 2hxy + by^2 = 0$ then prove that $\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a+b}\right|$ (3)

**Q.18.** Find the direction ratios of a vector perpendicular to the two lines whose direction ratios are $-2, 1, -1$ and $-3, -4, 1$. (3)

**Q.19.** Find the shortest distance between lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$. (3)

**Q.20.** Lines $\bar{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(2\hat{i} - 2\hat{j} + \hat{k})$ and $\bar{r} = (4\hat{i} - 3\hat{j} + 2\hat{k}) + \mu(\hat{i} - 2\hat{j} + 2\hat{k})$ are coplanar. Find the equation of the plane determined by them. (3)

**Q.21.** If $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + ..... + \infty}}}$, then

show that $\frac{dy}{dx} = \frac{\sec^2 x}{2y-1}$.

Find $\frac{dy}{dx}$ at $x = 0$. (3)

**Q.22.** Find the approximate value of $\sin(30°30')$.
Give that $1° = 0.0175^c$ and $\cos 30° = 0.866$ (3)

**Q.23.** Evaluate $\int x\tan^{-1} x\,dx$ (3)

**Q.24.** Find the particular solution of the differential equation $\frac{dy}{dx} = e^{2y}\cos x$, when $x = \frac{\pi}{6}$, $y = 0$ (3)

**Q.25.** For the following probability density function of a random variable X, find (a) P(X < 1) and (b) P(| X| < 1).

$f(x) = \frac{x+2}{18}$ ; for $-2 < x < 4$

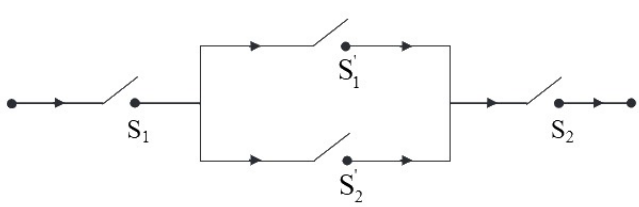
$= 0$ , otherwise (3)

**Q.26.** A die is thrown 6 times. If 'getting an odd number' is a success, find the probability of at least 5 successes. (3)

## SECTION – D

**Attempt any FIVE of the following questions:** **[20]**

**Q.27.** Simplify the given circuit by writing its logical expression. Also write your conclusion.




(4)

**Q.28.** If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ verify that $A(adjA) = (adjA)A = |A|I$ (4)

**Q.29.** Prove that the volume of a tetrahedron with coterminus edges $\bar{a}$, $\bar{b}$ and $\bar{c}$ is $\frac{1}{6}\left[\bar{a}\ \bar{b}\ \bar{c}\right]$.

Hence, find the volume of tetrahedron whose coterminus edges are $\bar{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\bar{b} = -\hat{i} + \hat{j} + 2\hat{k}$ and $\bar{c} = 2\hat{i} + \hat{j} + 4\hat{k}$. (4)

**Q.30.** Find the length of the perpendicular drawn from the point P(3, 2, 1) to the line $\bar{r} = \left(7\hat{i} + 7\hat{j} + 6\hat{k}\right) + \lambda\left(-2\hat{i} + 2\hat{j} + 3\hat{k}\right)$ (4)

**Q.31.** If $y = \cos(m \cos^{-1} x)$ then show that $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + m^2y = 0$ (4)

**Q.32.** Verify Lagrange's mean value theorem for the function $f(x) = \sqrt{x+4}$ on the interval [0, 5]. (4)

**Q.33.** Evaluate: $\int \frac{2x^2 - 3}{(x^2 - 5)(x^2 + 4)} dx$ (4)

**Q.34.** Prove that: $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a - x)dx$ (4)